

Linguistic Distribution and Infrastructure Cost Analysis

Data Science & Infrastructure Report

August 21, 2026

1 Dataset Overview

The analysis was performed on a Bengali text corpus extracted from a CSV file. The corpus contains a substantial volume of text, allowing for robust statistical modeling of linguistic properties.

- **Total word count (N):** 17,586,583
- **Unique word count (V):** 607,534

2 Zipf's Law Analysis

Zipf's Law posits that the frequency of any word in a naturally occurring corpus is inversely proportional to its rank in the frequency table ($f \propto \frac{1}{r^\alpha}$).

2.1 Statistical Results

The dataset was transformed using logarithmic scaling to fit a linear regression model. The results confirm a strong adherence to a power-law distribution:

- **Alpha value (Slope):** 1.278747
- **R^2 (Fit accuracy):** 0.9654

Conclusion: With an R^2 score of 0.9654, the data demonstrates a highly accurate fit. **Zipf's Law is supported**, proving a strong power-law distribution within the Bengali corpus.

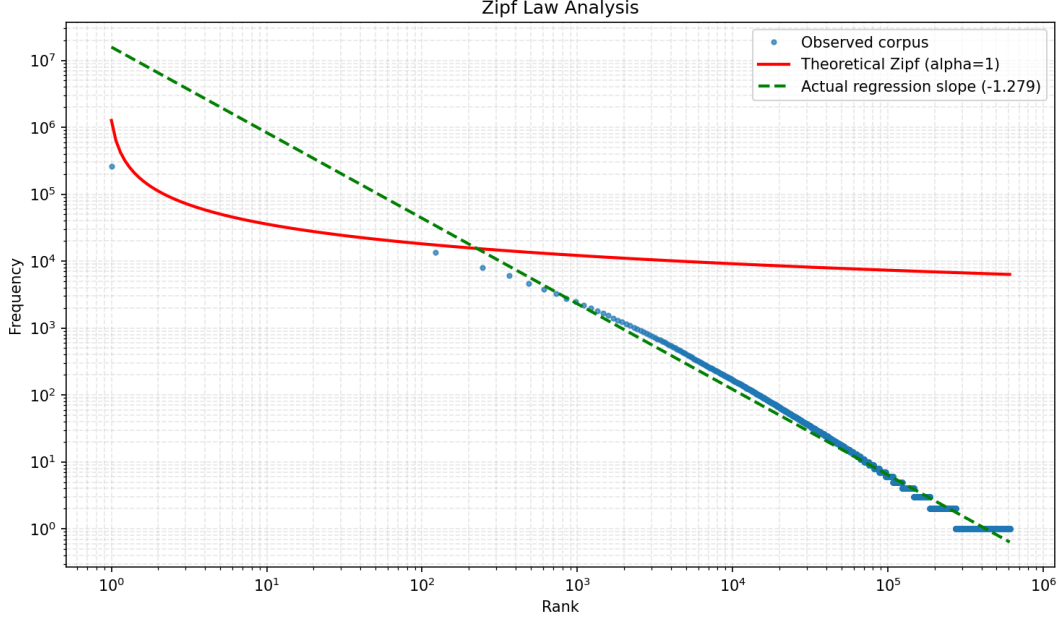


Figure 1: Log-log plot demonstrating the observed word frequencies against the theoretical Zipf curve.

3 Heaps' Law Analysis

Heaps' Law describes vocabulary growth as a function of document length, modeled by the equation $V = K \cdot N^\beta$.

3.1 Statistical Results

- **K parameter:** 10.9928
- **Beta parameter:** 0.6534
- **R^2 (Fit accuracy):** 0.9995

3.2 Artifact Investigation

During the chunking process (10,000 words per chunk), an anomaly or “artifact” was detected at the initial cumulative $N = 10,000$. The chunk introduced a massive spike of 3,974 new unique words.

Sample of new words triggering the artifact: বংশী, বাংলাদেশে, একটি, কৃষ্ণদ্রতম, উপজাতি, এরা, টাঙগাইল, জলোর, মহানন্দপুর, এবং, দন্দোনিয়া, নামে, পাশাপাশি, দুইটি, গ্রামরে

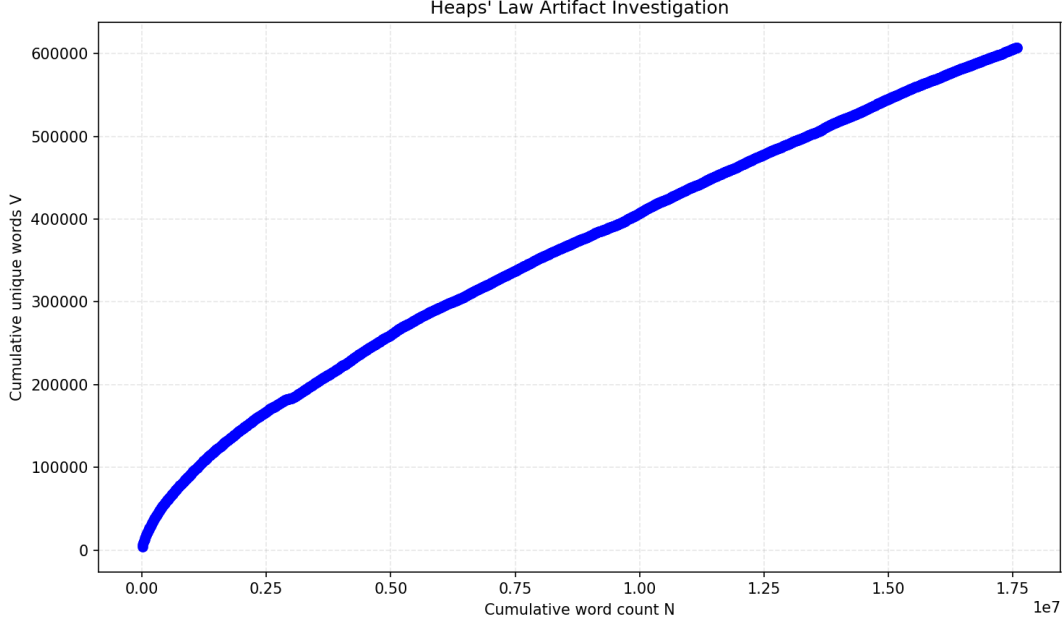


Figure 2: Cumulative unique words (V) plotted against total word count (N).

4 Infrastructure & Investment Cost Estimation

Based on an extrapolated computational baseline of \$1,000 per 100,000 words processed, the following theoretical cost estimations highlight the divergent business models of Generative AI versus traditional Search Retrieval systems.

Scenario	Estimated Cost (USD)
Current CSV file processing	\$175,865.83
Sarvam AI (Generative LLM)	\$40,594,000.00
Google Search (Retrieval)	\$20,000,000.00

Table 1: Comparative Infrastructure Costs

4.1 Cost Reasoning & Architectural Differences

Generative LLM (e.g., Sarvam AI) The massive \$40.5M estimated cost is heavily front-loaded into **data acquisition and model training**. LLMs require billions of words of training data (assumed here at 4 billion). The system does not merely store this text; it mathematically processes every token thousands of times across clusters of expensive GPU hardware (e.g., NVIDIA H100s) to learn linguistic patterns. While the upfront compute and engineering costs are astronomical, the ongoing storage and deployment costs post-training are comparatively smaller.

Search Engine Retrieval (e.g., Google) The \$20M search estimate reflects a fundamentally different architecture. Search engines require less upfront data manipulation (modeled here at 200 million words) because they index rather than train. However, their

ongoing infrastructure costs are massive. To achieve global, sub-second latency, retrieval systems require continuous web-crawling compute, 24/7 uptime across thousands of edge-caching servers, and massive global load balancers.